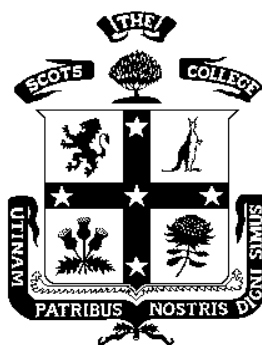

THE SCOTS COLLEGE



MATHEMATICS EXTENSION I

YEAR 12 ASSESSMENT TASK 3

17TH JUNE 2013

GENERAL INSTRUCTIONS

- Reading time – 2 minutes
- Working time – 45 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- Show all necessary working in Section II

WEIGHTING

20%

TOTAL MARKS

26

QUESTION 1 – 4 (MARKS AS INDICATED)

- Answers to be recorded in the answer booklets provided.
- *Each question must be completed in a new answer booklet.*
- Label each answer booklet with your name and teachers name and the question number attempted.
- Clearly indicate the booklet order if more than one booklet is used for a question. Eg) Book 1 of 2 and 2 of 2.

QUESTION 1**SEQUENCES, SERIES AND PROBABILITY****5 MARKS**

Adam plays a game of dice for a single player with two, 6-sided dice, numbers on each face with the numerals 1 to 6. He wins if the sum of the two upper most faces on each dice after a role is 4, 6 or 8. Conversely, if he roles a sum of 2, 3, 5 or 7 he records a loss. Any other sum of the two dice permits him to continue playing until he records a win or a loss.

- a) Find the probability that Adam wins on his first throw of the dice; 1 mark
- b) Show that the probability of Adam throwing the dice again is $\frac{5}{18}$; 1 mark
- c) Show that the probability that Adam wins on the second throw is $\frac{5}{18} \times \frac{13}{36}$; 1 mark
- d) Find the probability that Adam wins the game eventually. 2 marks

QUESTION 2**SEQUENCES AND SERIES****8 MARKS**

- a) A spherical bubble is expanding so that its volume is increasing at $10 \text{ mm}^3\text{s}^{-1}$. At what rate is the radius increasing when the surface area is 500 mm^2 ? 3 marks
- b) The rate at which a body warms in air is proportional to the difference between its temperature B and the constant temperature B_o of the surrounding air, i.e.,
$$\frac{dB}{dt} = k(B - B_o)$$

Where t is the time in minutes and k is a constant.
- i) Show that $B = B_o + Ae^{kt}$, where A is a constant, is a solution of $\frac{dB}{dt} = k(B - B_o)$ 1 mark
- ii) For a particular body, when $t = 0$, $B = 5^\circ\text{C}$ and when $t = 20$, $B = 15^\circ\text{C}$. Given $B_o = 25^\circ\text{C}$, find the temperature of the body after a further 30 minutes have elapsed. Give your answer to the nearest degree. 3 marks
- iii) Explain the behaviour of B as t becomes large. 1 mark

QUESTION 3

MATHEMATICAL INDUCTION

4 MARKS

Prove by Mathematical Induction that $3n^3 + 6n$ is divisible by 9 for all integers $n \geq 1$.

4 marks

QUESTION 4

APPLICATIONS OF CALCULUS TO THE
PHYSICAL WORLD

9 MARKS

An amount $\$A$ is borrowed at $r\%$ per annum reducible interest, calculated monthly. The loan is to be repaid in equal monthly instalments of $\$M$.

Let $R = \left(1 + \frac{r}{1200}\right)$ and let $\$B_n$ be the amount owing after n monthly repayments have been made.

- a) Show that $B_1 = AR - M$ 1 mark
- b) Show that $B_3 = AR^3 - M(R^2 + R + 1)$ 1 mark
- c) Show that $B_n = AR^n - M\left(\frac{R^n - 1}{R - 1}\right)$ 2 marks

Julie borrows $\$900\,000$ at 6% per annum reducible interest, calculated monthly. The loan is to be repaid in 360 equal monthly instalments.

- d) Using the information provided in Parts a to c, show that the monthly instalment should be $\$5395.95$ 2 marks
- e) With the 60th repayment, Julie pays an additional $\$120\,000$ so this payment is $\$125\,395.95$. After this, repayments continue at $\$5395.95$ per month. How many more repayments will be needed to pay off the loan? 3 marks

END OF EXAM.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

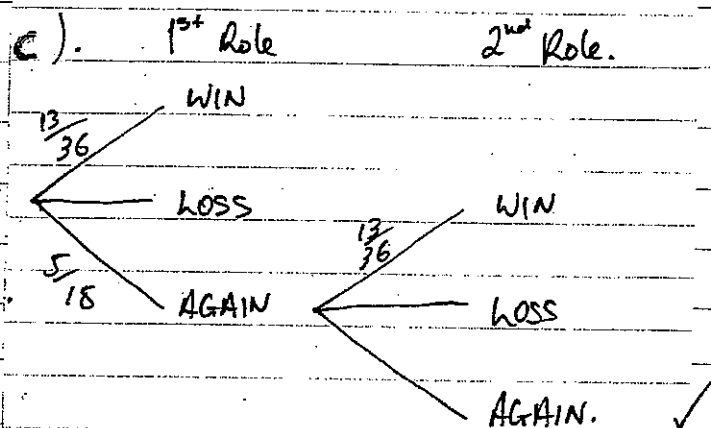
17th JUNE 2013

1.)

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

a). $P(\text{win}) = \frac{13}{36}$ ✓

b). $P(\text{throwing again}) = P(9, 10, 11 \text{ or } 12)$
 $= \frac{10}{36}$
 $= \frac{5}{18}$ ✓



$\therefore P(\text{win 2nd role}) = P(\text{AGAIN}) \text{ and } P(\text{WIN})$
 $= \frac{5}{18} \times \frac{13}{36}$ ✓

d. $P(\text{win eventually}) = P(\text{win 1st role}) + \dots$
 $+ P(\text{win 2nd role}) + P(\text{win 3rd role}) + \dots$
 $+ \dots + P(\text{win } n^{\text{th}} \text{ role})$

$= \frac{13}{36} + \left(\frac{5}{18} \times \frac{13}{36} \right) + \left(\frac{5}{18} \times \frac{5}{18} \times \frac{13}{36} \right) + \left(\frac{5}{18} \times \frac{5}{18} \times \frac{5}{18} \times \frac{13}{36} \right) + \dots$
 $+ \dots + \left(\frac{5}{18} \right)^n \times \frac{13}{36}$

$= \frac{13}{36} \left(1 + \frac{5}{18} + \left(\frac{5}{18} \right)^2 + \left(\frac{5}{18} \right)^3 + \dots + \left(\frac{5}{18} \right)^n \right)$
 $= \frac{13}{36} \left(\frac{1}{1 - \frac{5}{18}} \right)$ ✓
 $= \frac{13}{36} \times \frac{18}{13}$
 $= \frac{1}{2}$ ✓

Question 2.

a). $\frac{dV}{dt} = 10 \text{ mm}^3 \text{ s}^{-1}$ $SA = 500 = 4\pi r^2$

$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

Now:

$V = \frac{4}{3}\pi r^3$ when $SA = 500$

$\therefore \frac{dV}{dr} = 4\pi r^2$ $r^2 = \frac{500}{4\pi} = \frac{125}{\pi}$

$\therefore 10 = 4\pi r^2 \cdot \frac{dr}{dt}$

$\therefore 10 = 4\pi \times \frac{125}{\pi} \cdot \frac{dr}{dt}$

$10 = 500 \cdot \frac{dr}{dt}$

$\therefore \frac{dr}{dt} = 0.02 \text{ mm s}^{-1}$

bi). Given $B = B_0 + Ae^{kt}$

$\frac{dB}{dt} = kAe^{kt}$ where $B - B_0 = Ae^{kt}$

$\therefore \frac{dB}{dt} = k(B - B_0)$ ✓

Hence $B = B_0 + Ae^{kt}$ is a solution of $\frac{dB}{dt} = k(B - B_0)$

bii). When $t=0$, $B=5$

$$\therefore 5 = 25 + Ae^0$$

$$\therefore A = -20$$

$$\text{So, } B = 25 - 20e^{kt}$$

$$\text{When } t=20, B=15$$

$$\therefore 15 = 25 - 20e^{20k}$$

$$20e^{20k} = 10$$

$$e^{20k} = \frac{1}{2}$$

$$20k = \log_e \frac{1}{2}$$

$$k = \frac{1}{20} \log_e \frac{1}{2}$$

$$k = -0.034657$$

$$\text{Then } B = 25 - 20e^{-0.034657t}$$

After a further 30 minutes i.e. $t=50$,

$$B = 25 - 20e^{-0.034657 \times 50}$$

$$B = 21^{\circ} \text{ (to the nearest degree).}$$

biii). As t becomes larger, the term $e^{kt} \rightarrow 0$ ($k < 0$) and B approaches

the limiting value given by

$B = 25^{\circ}\text{C}$, i.e. the body acquires the temperature of the surrounding air.

Question 3

$$\text{a). LHS } S_n = (1 \times 4) + (2 \times 5) + (3 \times 6) + \dots + n(n+3)$$

$$\text{RTP } S_n = \frac{1}{3}(n+1)(n+5)$$

For $n=1$

$$\text{LHS} = 1 \times 4 \quad \text{RHS} = \frac{1}{3}(1+1)(1+5)$$

$$= 4 \quad = \frac{1}{3}(2 \times 6)$$

$$= 4$$

So the statement is true for $n=1$. Assume that it is true for $n=k$.

$$\therefore S_k = \frac{1}{3}(k+1)(k+5)$$

add the next term

$$T_{k+1} = (k+1)[(k+1)+3]$$

$$= (k+1)(k+4)$$

$$\text{So, } S_{k+1} = S_k + T_{k+1}$$

$$= \frac{k}{3}(k+1)(k+5) + (k+1)(k+4)$$

$$= (k+1)\left[\frac{k}{3}(k+5) + (k+4)\right]$$

$$= (k+1)\left[\frac{k^2}{3} + \frac{5k}{3} + \frac{3k}{3} + \frac{12}{3}\right]$$

$$= \frac{k+1}{3}(k^2 + 8k + 12)$$

$$= \frac{k+1}{3}(k+2)(k+6)$$

$$= \frac{k+1}{3}[(k+1)+1][(k+1)+5]$$

Hence, the statement is true for $n=k+1$. By the principle of Mathematical Induction it is true for all $n \geq 1$.

$$\text{b). } S_n = 3n^3 + 6n$$

RTP S_n is divisible by 9.
for $n=1$

$$S_1 = 3(1)^3 + 6(1)$$

$$= 3 + 6$$

$$= 9$$

$\therefore S_1$ is divisible by 9

Assume that S_n is divisible by 9 when $n=k$;

$$\therefore S_k = 3k^3 + 6k = 9M$$

$$S_{k+1} = 3(k+1)^3 + 6(k+1)$$

$$= 3(k^3 + 3k^2 + 3k + 1) + 6k + 6$$

$$= 3k^3 + 9k^2 + 9k + 3 + 6k + 6$$

$$= 3k^3 + 9k^2 + 9k + 6k + 9$$

$$= 3k^3 + 6k + 9k^2 + 9k + 9$$

$$= 3k^3 + 6k + 9(k^2 + 2 + 1)$$

$$= 9M + 9(k^2 + k + 1)$$

$\therefore S_{k+1}$ is divisible by 7. Hence, by the Principle of Mathematical Induction the Statement is true of for all $n \geq 1$.

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Question 4.

a). $B_1 = A \times \left(1 \times \frac{r}{100} \times \frac{1}{12}\right) - M$

$B_1 = A \times \left(1 \times \frac{r}{1000}\right) - M$ ✓

$B_1 = AR - M$

b). $B_2 = B_1 \times R - M$

$B_2 = (AR - M)R - M$

$B_2 = AR^2 - MR - M$

$B_3 = B_2 \times R - M$

$B_3 = (AR^2 - MR - M)R - M$

$B_3 = AR^3 - MR^2 - MR - M$

$B_3 = AR^3 - M(R^2 + R + 1)$

c). $\therefore B_n = AR^n - M(1 + R + R^2 + R^3 + \dots + R^{n-1})$

$B_n = AR^n - M(\text{sum of a GP})$

in GP, common ratio $R > 1$ ✓

$\therefore B_n = AR^n - M\left(\frac{R^n - 1}{R - 1}\right)$

d). $B_{360} = 0$

$0 = 900000(R^{360}) - M\left(\frac{R^{360} - 1}{R - 1}\right)$

where $R = \left(1 + \frac{r}{1200}\right)$

$R = \left(1 + \frac{5}{1200}\right)$

$R = 1.005$

$\therefore 0 = 900000(1.005^{360}) - M\left(\frac{1.005^{360} - 1}{1.005 - 1}\right)$

$M\left(\frac{5.022575212\dots}{0.005}\right) = 5420317.691\dots$

$M(1004.515042\dots) = 5420317.691\dots$

$M = \$5395.95$

e). Amount owing after 60 payments is $B_{60} - 120000$

$B_{60} = 900000(1.005)^{60} - 5395.95\left(\frac{1.005^{60} - 1}{0.005}\right)$

$= 1213965.187\dots - 5395.95(69.77003\dots)$

$= 1213965.137 - 376475.5961$

$= \$837489.54$

$\therefore$ Amount owing after 60 payments is:

$837489.54 - 120000 = \$717489.5412$

If payments continue after this at \$5395.95 per month:

$A = \$717489.54$

$M = \$5395.95$

and when the loan is repaid

$B_n = 0$

$\therefore 717489.54(1.005)^n - 5395.95\left(\frac{1.005^n - 1}{0.005}\right) = 0$

$717489.54(1.005)^n - \frac{5395.95(1.005)^n}{0.005} + \frac{5395.95}{0.005} = 0$

$(1.005)^n = -\frac{5395.95}{0.005} \div \frac{717489.54 - 5395.95}{0.005}$

$(1.005)^n = 2.983656698\dots$

$n \log_e 1.005 = \log_e 2.983656698\dots$

$n = \frac{\log_e 2.983656698\dots}{\log_e 1.005}$

$\log_e 1.005$

$n = 219.18$

There will be at most 220 further payments needed.

$n = 219.1760461\dots$

$n \hat{=} 219$ (accept this answer).

